

§1.582 — regularity as a pair of Horn sentences

Reading of Freyd & Scedrov, *Categories and Allegories* §1.582. Composition in **diagram order**: xy means first x then y . Companion to `Fredy/S1_58.lean` (`image_via_coeq`, §1.581 `bicart_repr_preserves_covers`, §1.583 `effectiveness_iff_coeq_pullback`).

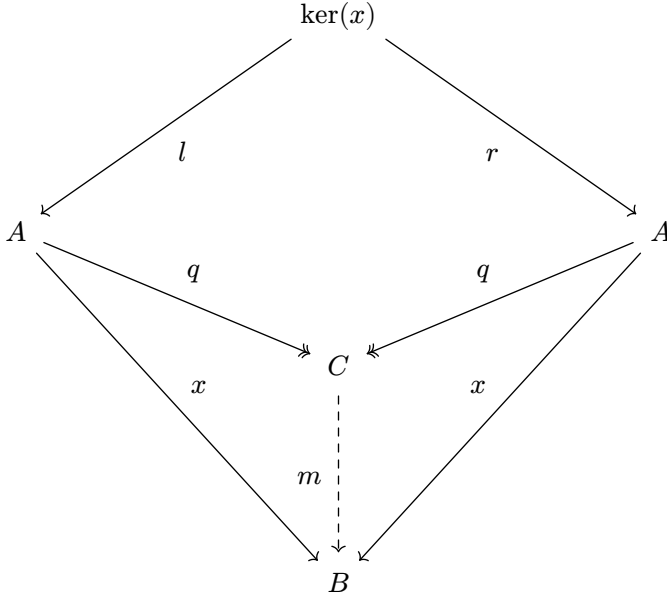
The claim

If $\mathcal{A}$ is **bicartesian** (Cartesian + coCartesian — has finite limits **and** colimits), then the predicate “ $\mathcal{A}$ is regular” is a **pair of Horn sentences in the bicartesian predicates**.

The reason: in a bicartesian category the **image** of a map is **built by operations**, not merely asserted to exist — so the existential quantifiers in “regular” disappear and what is left is purely universal equational implications.

The reason, in one diagram

Two nested diamonds sharing the apex $\ker(x)$ and the two copies of A — the level, its coequalizer, and the induced m :



Outer diamond $\ker(x), A, A, B$ — the **level** (kernel pair) of x , a pullback; it commutes, so $l; x = r; x$. **Inner diamond** $\ker(x), A, A, C$ — the **coequalizer** q of l, r (a cover); $l; q = r; q$. Since x collapses l, r too, the coequalizer’s universal property yields a **unique** $m : C \rightarrow B$ with $q; m = x$ — drawn **dashed**.

Every piece is an **operation** (pullback, coequalizer), never an “there exists” — which is exactly what lets regularity be **Horn**. (That m is moreover **monic** is the first Horn sentence, below.)

What “Horn sentence” means here

A **Horn sentence** is $\forall \dots, (e_1 \wedge \dots \wedge e_k) \implies e_0$ — a conjunction of equations implying one equation. The load-bearing feature is **no** $\exists$: every witness is supplied by a function symbol (the bicartesian operations). Stated naïvely, “regular” has existentials (“there **exists** an image”, “there **exists** a lift through a pullback”); Skolemizing them against the operations turns each into Horn form.

The pair of Horn sentences

(H1) The induced m is monic

The constructed factorization really is **cover-then-mono**, so images exist:

$$W \begin{array}{c} \xrightarrow{u} \\ \xrightarrow{v} \end{array} C \rightharpoonup \xrightarrow{m} B$$

$\forall u \, v, \quad u; m = v; m \implies u = v$. A single Horn sentence — m is the **operationally defined** map, so there is no $\exists$.

In a general bicartesian category m need **not** be monic, so requiring it is genuine content. This is exactly `image_via_coeq` (`Fredy/S1_58.lean`): `Mono (hcoeq.desc x kp_sq)`.

(H2) Covers are stable under pullback

The cover q , pulled back along **any** $h : D \rightarrow C$, is again a cover:

$$\begin{array}{ccc} P & \xrightarrow{\quad} & A \\ q' \downarrow \lrcorner & & \downarrow q \\ D & \xrightarrow{h} & C \end{array}$$

Form the **pullback** of q along h (an operation); assert the leg q' is again a coequalizer (cover). Equational, hence Horn — no $\exists$, the pullback is constructed.

Putting it together

(H1) + (H2) are Barr’s two conditions for a regular category — **image factorizations exist** and **their cover-parts are pullback-stable** — now both phrased with operations instead of $\exists$. That is the “pair of Horn sentences in the bicartesian predicates”.

Horn-ness is the payoff property: it travels along structure-preserving functors and into products/substructures. Hence

- **§1.581** — a representation preserving the bicartesian operations automatically preserves regularity (`bicart_repr_preserves_covers`); and
- **§1.583** — effectiveness of an equivalence relation likewise collapses to one Horn condition, “the coequalizer square is a pullback” (`effectiveness_iff_coeq_pullback`).

In the language of diagrams (§1.39)

Freyd’s **language of diagrams** (§1.39) is a sequence of diagrams joined by $\forall/\exists$ — each $\exists$ -step **extending** the previous with new vertices/arrows, commutativity assumed except at a puncture mark, objects never identified (§1.391, so the expressible properties are exactly the equivalence-invariant ones). With the **primitive predicates** “this diagram is a **pullback** / **equalizer** / **product** / **terminator**” (§1.443–§1.444) it is moreover a genuine **deductive** system, and its **inference rules are the universal properties**:

- **pullback** — a commuting cone yields a unique mediator;
- **equalizer** — a map that equalizes factors uniquely through it;
- **mono** — right-cancellation ($um = vm \implies u = v$); dually **coequalizer** / **cover** — left-cancellation.

Diagram chasing is derivation in this system. (Soundness, §1.444: a Horn sentence over these predicates true in **Set** holds in **every** Cartesian category — so such a goal may even be discharged by chasing in **Set**.) So §1.582 is statable **and** provable here: below is (H1) as a $\forall$ -sentence, then a chase each of whose steps is one of these rules.

(H1) stated diagrammatically

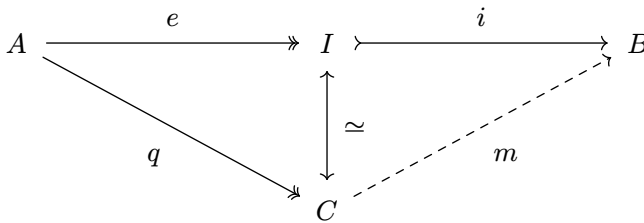
“ m is monic” is the $\forall$ -sentence: for the commuting diagram on the left (the puncture lets $u \neq v$), the two arrows must coincide —

$$\forall \quad W \begin{array}{c} \xrightarrow{u} \\ \xrightarrow{v} \end{array} C \rightharpoonup \xrightarrow{m} B \quad \text{with } u; m = v; m \implies u = v.$$

No $\exists$, no new object — a pure Horn implication between equations.

The chase: m is monic because C is the image of x

Factor x through its **image** (regularity: cover e then mono i). Two covers out of A — e and q — with monic / induced tails:



1. $l; x = r; x$, and i monic **right-cancels** [mono rule] to give $l; e = r; e$; so e coequalizes l, r , and the **coequalizer UMP** of q [coequalizer rule] yields a unique $\varphi : C \rightarrow I$ with $q; \varphi = e$.
2. The image cover e has kernel pair $\ker(x)$ (i monic) and **is** the coequalizer of it; so l, r coequalize through e , and the **coequalizer UMP** of e [coequalizer rule] yields a unique $\psi : I \rightarrow C$ with $e; \psi = q$.
3. $q; (\varphi; \psi) = e; \psi = q; \psi = q; 1$, and q a cover is **epic**, so it **left-cancels** [cover rule]: $\varphi; \psi = 1$; symmetrically $\psi; \varphi = 1$. So $C \simeq I$.
4. Under that iso m is the image mono i (both are the unique tail with $q; (-) = x = e; i$); iso ; mono is mono [composition]. So **m is monic**. ■

Every step is a rule of the diagram language — mono/cover cancellation and the coequalizer universal property. The two **regular** facts it stands on (an image factorization exists; a cover is the coequalizer of its kernel pair) are the standing predicates the rules act on. So this is a **derivation**, not an informal picture — and by §1.444 the same Horn goal could instead be checked in **Set** (where C is literally the set-image of x) and transported. In Lean it is `image_via_coeq` (`Fredy/S1_58.lean`), whose proof runs exactly these steps (`image.lift`, `cover_is_coequalizer_of_level`, `cover_epi`).